Breaking Myths: The Reality of Galaxy's Capabilities and Impact

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Open science computing is vast and evolving, making workflow managers like Galaxy essential for reproducible and accessible data analysis.
Established in 2005, **Galaxy** supports diverse scientific domains and is widely adopted in France, with 10+ servers and UseGalaxy.fr as the national instance.

Despite strong adoption and training resources, misconceptions persist, such as Galaxy being only for genomics or being difficult to use efficiently for "real" analysis. To address this, the Galaxy Community works actively on **debunking the misconceptions**.

1

Users & Educators
"Galaxy is only useful for genome scientists"

Galaxy is supporting all omics-related research and goes beyond Life-Science.

Metabolomics, Climate, Natural Language Processing, Astronomy, Material Science, ML & AI, etc.

2

Data Scientists
"Galaxy offers nothing to technically capable informaticians who can write their own analysis code"

Galaxy actually offers numerous benefits and features that can enhance the productivity and efficiency of technically capable users.

Workflow Manager, Interactive Tools (Jupyter, etc), OpenAPI, AI Models, No hassle to install tools (and their dependency) or submit jobs to clusters.

3

Users & Educators
"Galaxy does not scale to large and complex problems"

"Yes it can!" - Someone who has actually used Galaxy

65,000+ jobs per day on the usegalaxy.eu, .org, .org.au, and .fr. servers; 90+ FAIR and peer-reviewed large-scale workflows; COVID-19, IRIDA, ABRomics data processing

4

Users & Educators
"Galaxy is hard to use."

"It's easier to use than the alternatives!" - A bench scientist

3 mechanisms: standardised environments; a User Interface / User Experience-focused working group with experts & community feedback; and extensive, FAIR training material to address any remaining obstacles.

5

Users & PIs
"Galaxy is only useful for teaching, it's not popular amongst scientists in academia or industry.""

Its user-friendly interface, scalability, and extensive toolset make it ideal for both education and complex research, as shown by its 500,000+ users, its use in high-impact studies -cited in 30+ publications every week- and industry applications.

6

Galaxy cannot be used on secured data

Galaxy is used for sensitive data, both in large scale genomic projects and in hospitals as it offers a flexible and scalable environment, leveraging security-enhanced deployment models and access controls.
Bring Your Own Compute/Storage/Data (BYOC/S/C), Role-based access control (RBAC), Single Sign-On (SSO), Identity Federation Protocols, etc.

7

Galaxy is free to use, so its quality must be poor because there are no professional staff members.

The Galaxy community is full of active and organised professionals, from software engineers to life scientists.

Combination of automation & testing, community structure, user engagement, One of the best scientific open source projects code quality wise.

8

Galaxy is open-source & open-development, so there cannot be good documentation or support.

The Galaxy project is extensively documented, with live support and training.

Online documentation; GitHub issues and PR, Release notes; Galaxy help forum; Galaxy Training Network with 400+ tutorials, 200+ videos

9

PIs & Core Service Decision-Makers
"Galaxy cannot be sustainable in the long run, because no commercial organisation is managing it."

Strong governance, diverse funding, user and developer community, and open-source model make it highly sustainable.

Project successfully sustained for the last 20 years. Evolving governance and administrative structures. Sustainability report.

10

Galaxy tools and supported analyses are out-dated

Galaxy's semi-automatic update infrastructure helps keep tools and workflows as current as possible while balancing stability and reproducibility.

~30 tool updates / month in the last 5 years via GitHub (CI) pipelines.

Share Your Insight!

If you have any misconceptions about Galaxy that haven't been addressed yet, or if you'd like to share additional examples that support its use, please contribute to this discussion. Take a moment to jot down your thoughts on one of the paper you can find below. Your input is invaluable in helping us debunk misconceptions around Galaxy.

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